

Cloud Infrastructures

01 - Cloud and Virtualisation

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CY Cergy Paris Université

ING3 - Ingénierie du Cloud Computing

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Table of Contents

- 1 Course Introduction
- 2 Cloud Computing : Introduction
- 3 Virtualisation
- 4 Service and Deployment models
- 5 Cloud providers and services
- 6 Drawbacks and challenges
- 7 Introduction to OpenNebula

Table of Contents

- 1 **Course Introduction**
- 2 Cloud Computing : Introduction
- 3 Virtualisation
- 4 Service and Deployment models
- 5 Cloud providers and services
- 6 Drawbacks and challenges
- 7 Introduction to OpenNebula

About me

Juan Ángel Lorenzo del Castillo



- B.Sc. in Telecom Engineering majoring Electronics from the University of Valladolid (Spain).
- M.Sc. in Telecom Engineering from the University of Valladolid (Spain).
- Ph.D. parallel and distributed systems from the University of Santiago de Compostela (Spain).



- Research engineer at the Cloud and Security Lab (Hewlett Packard Labs, Bristol).
- Associate professor at CY-Tech. Co-director of the *High Performance Data Analytics* option.
- Researcher at ETIS Laboratory (CY Cergy Paris Université).
- Research interests:
 - Cloud computing infrastructures.
 - Operating systems.
 - Performance monitoring of large-scale architectures.
 - Data profiling in real and virtualised environments.
 - Scheduling techniques in the IoT-Edge-Cloud continuum using Reinforcement Learning.

About you

- Who are you?
- What is your background?
 - ▷ What do you know about Distributed Systems?
 - ▷ Do you have any computing interests outside the school?
 - ▷ Geek-o-meter:
 - Have you ever installed and fully configured an operating system?
 - Have you participated in any Open Source project?
 - Have you ever compiled the Linux kernel?
- What is Cloud Computing for you?
- What do you expect from this course?
- **Requirements:**
 - ▷ A *@cy-tech.fr* or *@etud.univ-pau.fr* email account.
 - ▷ A laptop running a Linux distribution.
 - ▷ Lots of free space in your hard drive.

Course outline

1. Introduction to Cloud Computing and virtualisation

- ▷ Definitions
- ▷ Types of Cloud infrastructures
- ▷ Hypervisors
- ▷ Cloud Platforms examples

2. Virtualisation with KVM

- ▷ Automated cluster installation

3. Performance monitoring (if time allows it)

- ▷ Perf events
- ▷ Zabbix
- ▷ Netdata

4. Introduction to Docker

5. Microservices with Docker

6. Container orchestration with Kubernetes

7. The Google Cloud Platform

- ▷ Kubernetes on the GCP

Pedagogy and Assessment

Pedagogy:

- This course will be taught in English.
- The course slides and additional materials will be either in French or English.
- *Constructivist approach*: I explain, I propose. You try by yourselves, you learn by experimenting.

Assessment:

- One single exam.
- Some exercises may be graded.

Table of Contents

- 1 Course Introduction
- 2 Cloud Computing : Introduction
- 3 Virtualisation
- 4 Service and Deployment models
- 5 Cloud providers and services
- 6 Drawbacks and challenges
- 7 Introduction to OpenNebula

What is the Cloud?

What is the Cloud?

You already are a cloud user



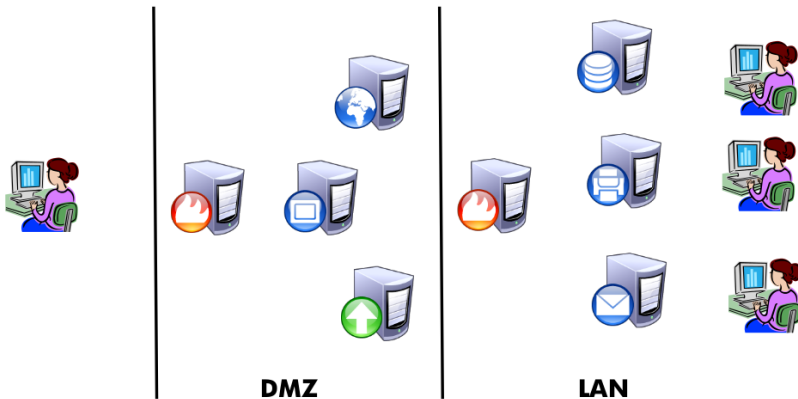
What is the Cloud?

Origins of the Cloud Computing

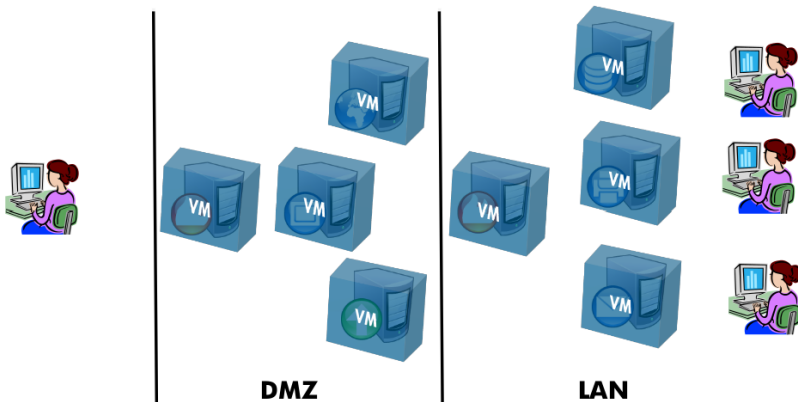
Need for Utility Computing
Underutilised resources
Evolution of the virtualisation technology

What is the Cloud?

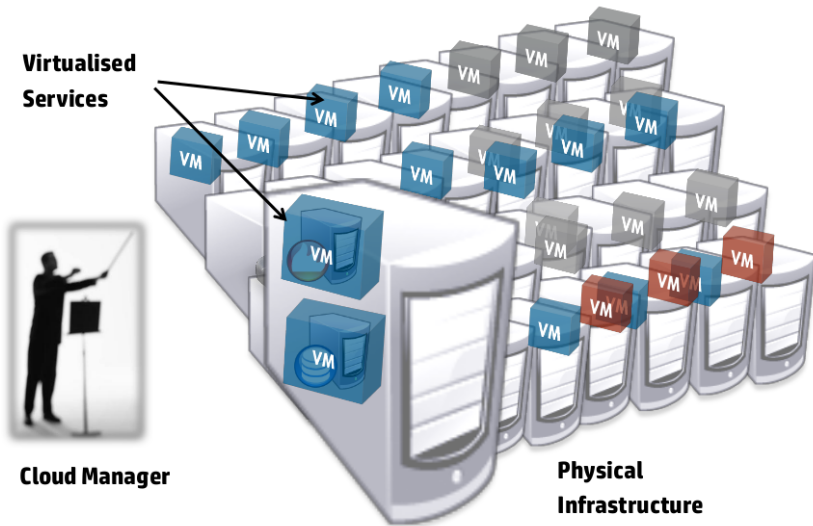
Traditional approach



What is the Cloud? Virtualised approach



What is the Cloud? Virtualised approach



What is the Cloud?

Cloud definition

Definition by NIST¹

Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction. This cloud model is composed of five essential characteristics, three service models, and four deployment models.

¹ <https://nvlpubs.nist.gov/nistpubs/Legacy/SP/nistspecialpublication800-145.pdf>

What is the Cloud?

Cloud definition

Definition by NIST (cont.)

Essential characteristics:

- **Ubiquitous network access:** resources available over Internet. This makes it accessible wherever Internet connectivity is available.
- **On-demand self-service:** consumer spontaneously asks for resources (storage/processing/memory/network bandwidth) without human intervention
- **Resource pooling:** Cloud providers have a large pool of computing resources (such as servers, storage, network capacity etc.) that is dynamically allocated to their customers on demand. The resources location is not specified or specified at a higher level (country, region, ...)
- **Rapid elasticity:** Capabilities can be elastically provisioned and released
- **Measured services:** Control and optimization of the resource usage by leveraging a metering capability

What is the Cloud?

Cloud definition

Essential characteristics:

				
On-demand self-service	Broad network access	Resource pooling	Rapid elasticity	Measured service
No human intervention needed to get resources	Access from anywhere	Provider shares resources to customers	Get more resources quickly as needed	Pay only for what you consume

What is the Cloud?

Cloud definition

Definition by Wikipedia²

Cloud computing is the on-demand availability of computer system resources, especially data storage and computing power, without direct active management by the user. The term is generally used to describe data centers available to many users over the Internet. Large clouds, predominant today, often have functions distributed over multiple locations from central servers. If the connection to the user is relatively close, it may be designated an edge server.

² https://en.wikipedia.org/wiki/Cloud_computing

What is the Cloud?

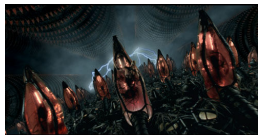
Cloud definition

Virtualisation
Scalability and Elasticity
Resources optimisation
Pay-per-use
SLAs (Service Level Agreement)

Table of Contents

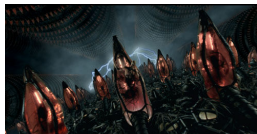
- 1 Course Introduction
- 2 Cloud Computing : Introduction
- 3 **Virtualisation**
- 4 Service and Deployment models
- 5 Cloud providers and services
- 6 Drawbacks and challenges
- 7 Introduction to OpenNebula

Virtualisation

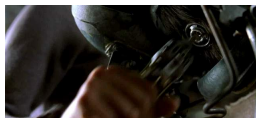


Hardware Infrastructure

Virtualisation

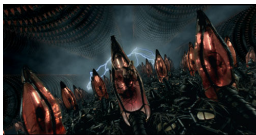


Hardware Infrastructure



Guest OS

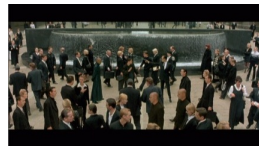
Virtualisation



Hardware Infrastructure

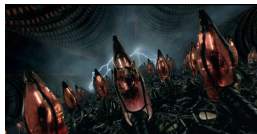


Guest OS

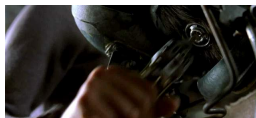


Virtual Environment

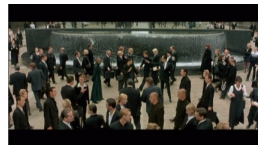
Virtualisation



Hardware Infrastructure



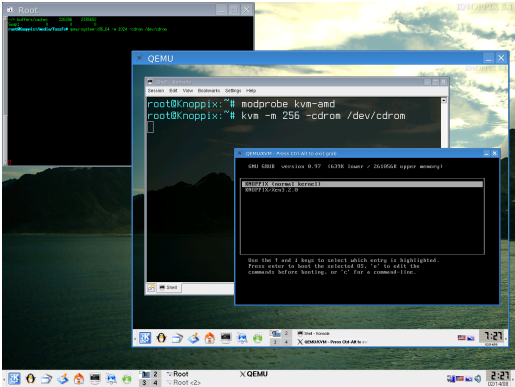
Guest OS



Virtual Environment

Virtualization refers to the act of creating a virtual (rather than actual) version of something, including virtual computer hardware platforms, storage devices, and computer network resources (Wikipedia).

Hypervisors

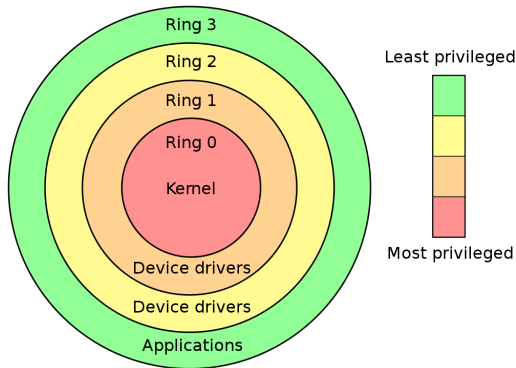


Virtualisation types

Binary Translation
Paravirtualisation
Hardware Virtualisation

Virtualisation types

The Privilege (or Protection) Rings

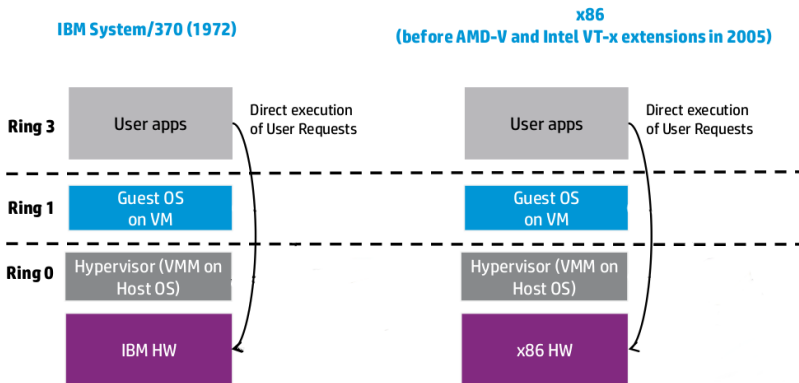


Privilege rings for the x86 available in protected mode (source: Wikipedia).

Virtualisation types

Ring Deprivileging in the hypervisor

The inability of the x86 architectures to virtualise

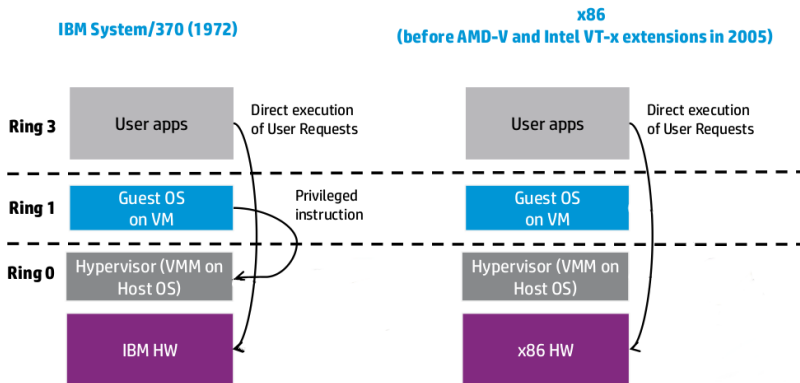


Ring deprivileging: the guest OSES are no longer running in ring 0, but with less rights in ring 1.

Virtualisation types

Ring Deprivileging in the hypervisor

The inability of the x86 architectures to virtualise

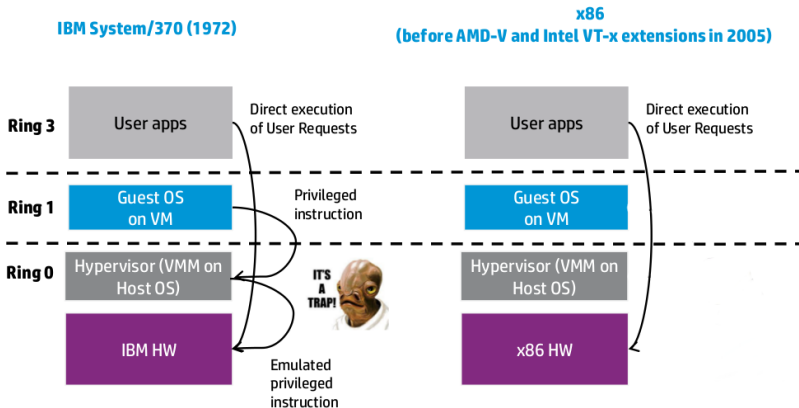


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Virtualisation types

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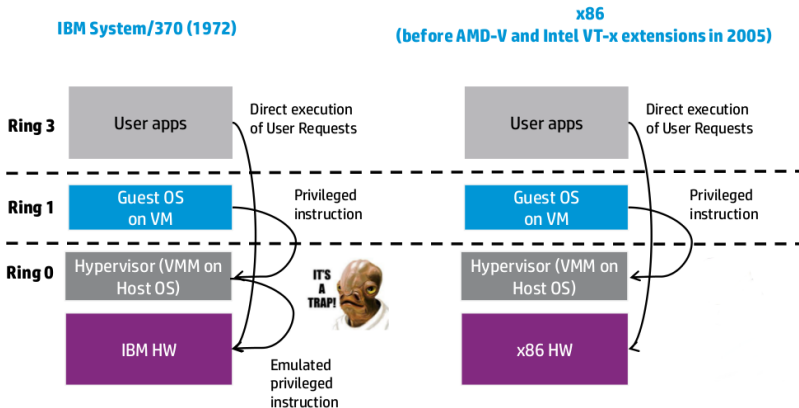


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Virtualisation types

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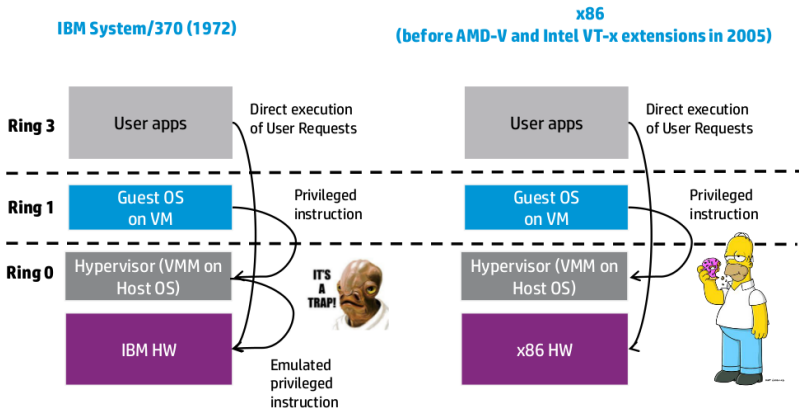


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Virtualisation types

Ring Deprivileging in the hypervisor

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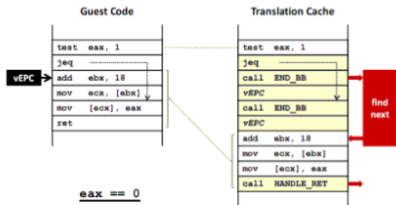
Virtualisation types

Binary Translation

Replacement of “critical” or “dangerous” code to safe code **in runtime**.

x86 (1999)

Controlling Control Flow



(Source: VmWare)

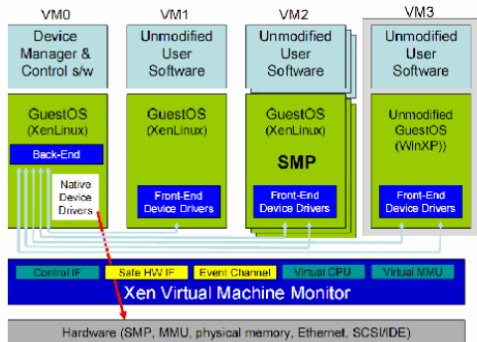


- Translation of the binary code that the kernel of a guest OS wants to execute on the fly
- User mode applications are assumed to be safe so they are executed directly as if they were running natively.

Virtualisation types

Paravirtualisation

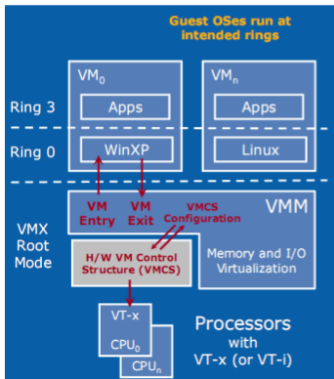
Replacement of “critical” or “dangerous” code to safe code **in the source code.**



(Source: Xen)

Virtualisation types

Hardware Virtualisation



(Source: VmWare)



- Fixes the problem that the x86 instructions architecture cannot be virtualised.
- Traps all exceptions and privileged instructions by forcing a transition from the guest OS to the VMM (*Virtual Machine Manager*): "VMexit".
- Can be considered an improved version of the IBM S/370 virtualization methods.
- Advantage: the guest OS runs at its intended privilege level (ring 0) and the VMM is running at a new ring with an even higher privilege level (Ring -1, or Root mode").
- Efficient: as long as system calls do not involve critical instructions, the guest OS can provide kernel services to the user applications.

Virtualisation types

Hypervisor classification

Type I (Bare Metal)



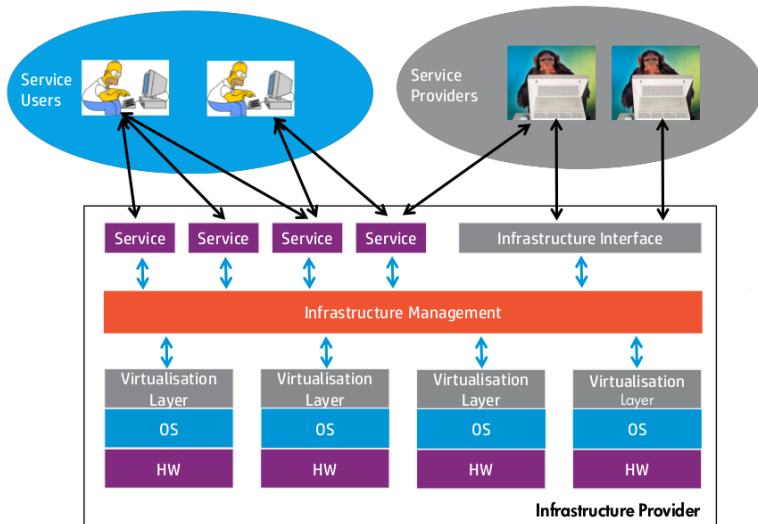
Type II



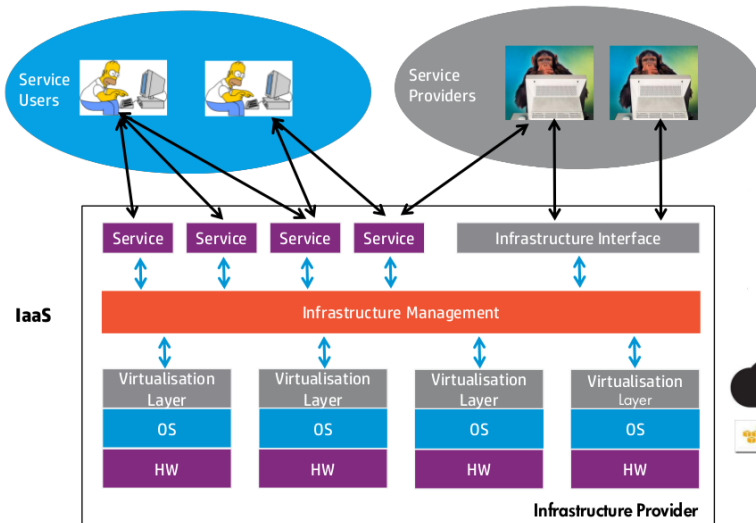
Table of Contents

- 1 Course Introduction
- 2 Cloud Computing : Introduction
- 3 Virtualisation
- 4 **Service and Deployment models**
- 5 Cloud providers and services
- 6 Drawbacks and challenges
- 7 Introduction to OpenNebula

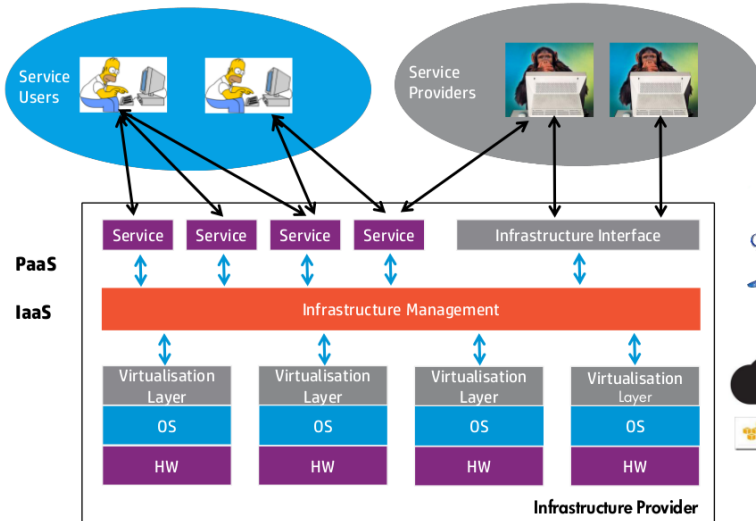
Cloud Actors - Service models



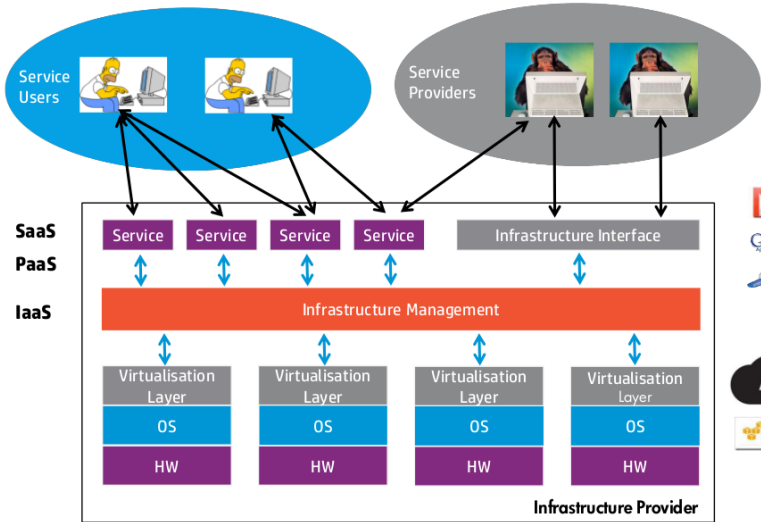
Cloud Actors - Service models



Cloud Actors - Service models

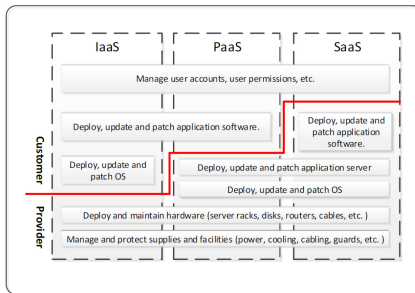


Cloud Actors - Service models



Cloud Service Models

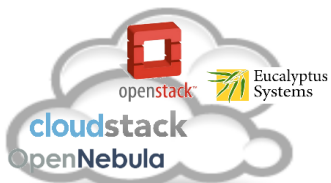
- **Infrastructure as a Service (IaaS):** the cloud service provider offers virtual machines, networking and storage capacities. Configuration on top of these bare resources is the responsibility of the cloud service user.
- **Platform as a Service (PaaS):** the cloud service provider offers a "Platform"(at different levels of abstraction). The provider is responsible for maintaining the platform, and the cloud service user can focus on deploying and maintaining their application(s). Examples: a LAMP stack or Google App Engine.
- **Software as a Service (SaaS):** moves the responsibility for the application itself to the cloud service provider. Examples: Microsoft Office 365 or GMail.



Source: Dr. M.A.C. Dekker: "Cloud Security Guide for SMEs"

Deployment models

Private Cloud



Public Cloud



Deployment models

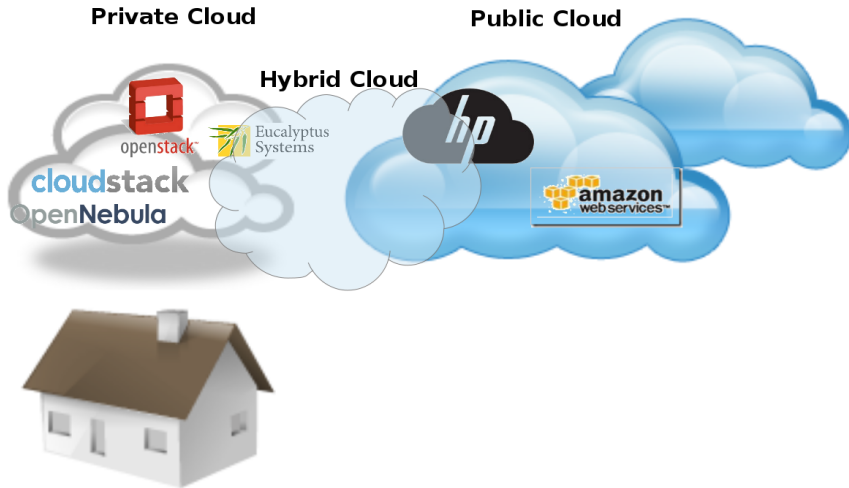


Table of Contents

- 1 Course Introduction
- 2 Cloud Computing : Introduction
- 3 Virtualisation
- 4 Service and Deployment models
- 5 **Cloud providers and services**
- 6 Drawbacks and challenges
- 7 Introduction to OpenNebula

Cloud providers and services

aws	Azure	Google Cloud	ORACLE CLOUD
Elastic Compute Cloud (EC2)	Virtual Machine	Compute Engine	Virtual Machine
Elastic Kubernetes Service (EKS)	Azure Kubernetes Service (AKS)	Google Kubernetes Engine (GKE)	Oracle Container Engine
Lambda	Azure Functions	Cloud Functions	OCI Functions
Simple Storage Service (S3)	Blob Storage	Cloud Storage	Object Storage
Elastic Block Store	Managed Disk	Persistent Disk	Persistent Volume
Elastic File System	File Storage	File Store	File Storage
Virtual Private Cloud	Virtual Network	Virtual Private Cloud	Virtual Cloud Network
Route 53	DNS	Cloud DNS	DNS
Elastic Load Balancing	Load Balancer	Cloud Load Balancing	Load Balancer
Web Application Firewall	Web Application Firewall	Cloud Armor	Web Application Firewall
RDS	SQL Database	Cloud SQL	ATP
DynamoDB	Cosmos DB	Firebase Realtime Database	NoSQL Database
Redshift	Synapse Analytics	BigQuery	Autonomous Data Warehouse
Elastic MapReduce	HDInsight	DataProc	Big Data
Kinesis	Streaming Analytics	Dataflow	Streaming
SageMaker	Machine Learning	Vertex AI	Data Science
Glue	Data Factory	Data Fusion	Data Integration
EventBridge	Event Grid	Eventarc	Events
Simple Queuing Service	Storage Queues	Pub/Sub	Streaming
Simple Notification Service	Service Bus	Firebase Cloud Messaging	Notifications
CloudWatch	Monitor	Cloud Monitoring	Monitoring
CloudFormation	Resource Manager	Deployment Manager	Resource Manager
IAM	Active Directory	Cloud Identity	IAM
KMS	Key Vault	Cloud KMS	Vault

Source: bytebytego

Table of Contents

- 1 Course Introduction
- 2 Cloud Computing : Introduction
- 3 Virtualisation
- 4 Service and Deployment models
- 5 Cloud providers and services
- 6 Drawbacks and challenges**
- 7 Introduction to OpenNebula

Drawbacks and challenges

- **Security and Privacy:** The user has to trust the cloud service provider.
 - ▷ Security: confidentiality + integrity + availability
 - ▷ Privacy: non-disclosure of information, for example.
- **Vendor Lock-in:** The difficulty of switching to a different provider.
- **Lack of Expertise:** Cloud computing is still relatively new, and the skills required depend on the deployment and service models used.
- **Performance Concerns:** Network limitations and performance implications of shared physical resources.
- **Organizational Issues:** For example, resistance to change.
- **Cost and Cost Management:** Cloud service providers typically offer IT resources at cheaper rates than smaller customer organizations can achieve when running their own IT. But it depends on the existing local infrastructure to migrate, the needs of the organisation and the volatility of costs in cloud computing.

Is it worth it? The hidden costs of public cloud platforms

Cost Type	Illustration	Hidden costs
Free Tier Ambiguity		AWS Free Tier has limits on many resources. Exceeding those limits can be costly.
Under-utilized Elastic IP (EIP) addresses		Free for 5. Any extra charged at hourly rate.
Unused Elastic Load Balancers		An unused Load Balancer is still charged at an hourly rate.
Unused Elastic Block Storage (EBS)		Unused EBS are charged at GB-month rate.
Orphan Elastic Block Storage Snapshots		Delete an EBS volume will NOT delete backups, create orphan backups cost.
S3 Access Charges		Get, List, and Retrieval request could be much more costly than file storage cost.
S3 Partial Uploads		Incomplete multipart uploads still incur charges
Transfer cost		Transfer to AWS is FREE, but transfer out can be costly.

Source: bytebytego

Organizational Issues...

- ... for CIOs (Chief Information Officers) I know the public cloud is good for business. But how can I take advantage of the vast array of cloud resources while retaining control and governance? How can I make public and hybrid cloud sourcing part of my IT strategy? And how can I possibly manage it all?
- ... for the Operations Team New roles. Operation teams will be freed up from the day-to-day logistics of managing the infrastructure and will instead get to focus on high-level problems such as automation, security, and analytics
- ... for Developers Would the decision to move an existing business application from dedicated in-house servers to a public cloud provider result in significant reworking of the program's code? New paradigm: Software Testing In The Cloud (STITC)
- ... for Users Flexibility to have resources on demand

Table of Contents

- 1 Course Introduction
- 2 Cloud Computing : Introduction
- 3 Virtualisation
- 4 Service and Deployment models
- 5 Cloud providers and services
- 6 Drawbacks and challenges
- 7 Introduction to OpenNebula

OpenNebula

Uniform management layer in charge of orchestrating different technologies.

Reference Users

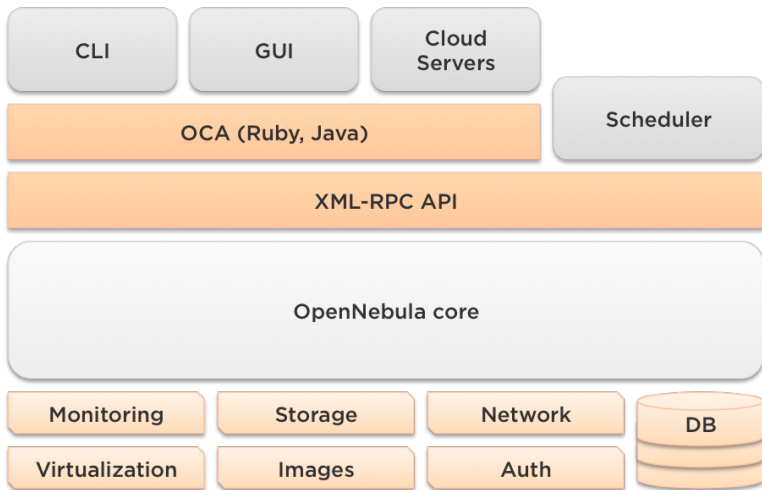


Properties

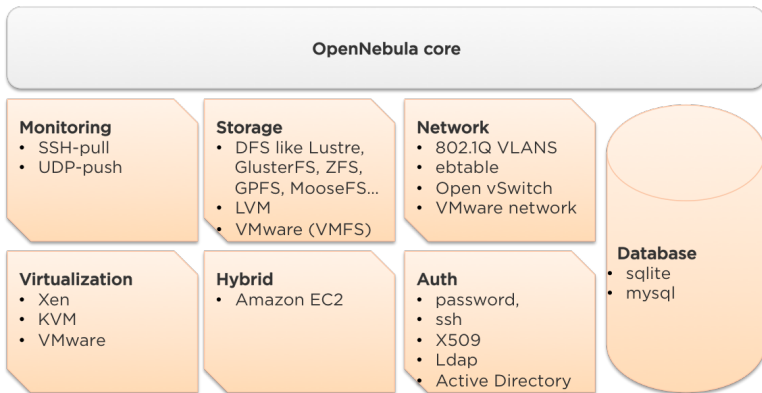
OpenNebula

- **Adaptability**: Capacity of integration adapted to all kinds of datacenter
- **Enterprise-ready** : Commercial Support and continuous updates
- **No Lock-in** : Platform independent and wide infrastructure
- **Light** : Efficient and simple
- **Tested** : Rigorously tested, mature, wide adoption
- **Powerful** : Advanced characteristics
- **Scalable** : Multi-layer architecture
- **Interoperable** : Rich APIs and interfaces
- **Open Source** : Apache v2 licence

Architecture



Architecture (II)



Basic Deployment

